Problem Statement and AI Sketch

Team: Karissa Kocourek and Daniel Escalante

In this assignment, each team will propose a tentative topic for their AI Design Project.

This brief proposal is focused on characterizing a specific domain problem to address (Problem Statement), and a non-technical description of how an AI tool or system might address the problem (AI Sketch).

Changes to this initial proposal are encouraged and expected as teams get further into the design and development process in the coming weeks.

Deadline: Wednesday, September 9 at 11:59 PM

Submission: Canvas *(All team members should submit the same file)*

Team Members

Add each team member to the table below.

- **Name:** List each team member's preferred name.
- **Email:** List each team member's UF email address (@ufl.edu).
- **Other Role(s)** (*Optional*)**:** In addition to being an AIBHS graduate student, list any other occupations or positions each team member currently holds, e.g., *RN*, *medical student*, *engineer,* etc. Full-time AIBHS students can enter *N/A*.
- **Coding Experience:** Indicate each team member's approximate level of coding experience with coding by selecting one of these options:
  - None
  - Low
  - Medium
  - High

  (Remember, no prior coding experience is required! This is for informational purposes.)

Name	Email	Other Role(s)	Coding
Karissa Kocourek	Withheld	Medical Lab Scientist	None
Daniel Escalante	Withheld	Independent Developer, Clinical AI Systems	High

Project Title

Enter a tentative, working project title below which identifies the general theme or direction of your project. You will likely change the title later as you refine the project.

Project Title:

Tracer — Predictive Specimen Tracking and Loss Prevention Across the Clinical-to-Laboratory Handoff

Problem Statement (2-3 paragraphs)

In 2-3 paragraphs, describe the specific, real-world problem or opportunity in biomedical or health sciences that your team wants to address.

Problems should be as concrete, specific, and detailed as possible. They should not be overly general or ambitious. The scope of projects should be narrow enough that a potential solution is feasible in just one semester.

Strong problem statements address the questions below. (*Note: these are only suggestions for guiding your problem statements. Do not answer them point-by-point.*)

- What is the specific problem? Who does it affect? How does the specific problem fit into a broader context?
- Why is it a problem? What are its negative consequences? What is the measurable impact of the problem?
- Why is the problem important and worth addressing? How would things improve if the problem were solved?
- What is currently known and/or unknown about the problem? What caused the problem?
- What is the ideal solution to the problem?
- What existing solutions attempt to address the problem (both non-AI and AI)? What are their limitations?
- What is the gap between the ideal solution and the current status quo? What is missing? What are the opportunities?

Problem Statement (2-3 paragraphs)

Between the moment a specimen is collected at a patient's bedside and the moment it is received into the clinical microbiology laboratory, no system knows where it is. The tests are ordered, the collection is documented, and the receipt at the laboratory is recorded. Everything in between is unobserved. For a specimen moving between floors, between buildings, or between campuses by pneumatic tube or courier, that stretch can run from a few minutes to several hours, and for its whole duration, the specimen has no recorded state. When a specimen never arrives, the absence surfaces indirectly and late. A clinician calls asking why a result has not been reported, or a technologist notices an unreceived specimen. This is the hardest failure in the workflow to catch, because a missing specimen produces no event, and every monitoring mechanism in the laboratory is triggered by something happening. Catching it requires reasoning about what should have arrived by now and has not.

The consequences are clinical and affect patient care. Specimen integrity degrades on a biological clock that runs at a different speed for every specimen type, and that clock does not pause while a tube sits in a transport bin. An unpreserved urine for culture held too long at room temperature overgrows with organisms picked up during collection. The colony count stops reflecting what was in the bladder, and the result can no longer separate infection from contamination. Guidance from the Infectious Diseases Society of America and the American Society for Microbiology therefore calls for boric acid preservative or refrigeration in transit (Miller et al., Clinical Infectious Diseases, 2018). Cerebrospinal fluid runs the opposite direction. The same guidance instructs that CSF be sent at room temperature immediately and states plainly that it must not be refrigerated, because the organisms causing bacterial meningitis lose viability in the cold. A specimen that is out of stability is rejected, and rejection means recollection. That is another needle stick, or another lumbar puncture, and hours added to a diagnosis while a patient stays on empiric antibiotics waiting for a result that no longer exists. Patients absorb the worst of it. Nurses and phlebotomists repeat collections, technologists field the calls and walk the route looking for tubes, and clinicians decide the next steps without test results. This is not a rare corner of laboratory medicine. In a hospital audit by Carraro and Plebani, 61.9% of confirmed testing errors originated before analysis, against 15% at the analytical bench (Clinical Chemistry, 2007). The bench has been automated and instrumented for decades. The corridor leading to it has not.

What exists today reports the past. Status screens in the laboratory information system show a specimen's last scan, which means they go blank for precisely the interval in question. Fixed turnaround-time rules raise a flag once a set number of minutes has elapsed, by which point the window has usually already closed, and a single threshold treats an emergency department spinal fluid at 2 AM the same as a clinic urine on a Tuesday afternoon. Retrospective quality reports count missed specimens a month later, when nothing can be recovered. Radio-frequency and Bluetooth tagging can deliver genuine physical location, at the cost of a tag on every specimen and readers through every corridor, which few laboratories will fund. When something does go wrong, the real response is a phone call and a person walking the route. Artificial intelligence has arrived in the clinical laboratory almost entirely downstream, interpreting images and validating results, and very little of it has been aimed at specimen logistics. The ideal system would know where every specimen is at all times and warn staff before one degrades or disappears. Hardware buys that visibility with capital expenditure most

laboratories will not approve. The gap is that nobody has tried to buy it with inference. The timestamps needed to estimate the unobserved interval are already being collected in routine operation, and they are simply never used that way.

AI Sketch (1-2 paragraphs)

In 1-2 paragraphs, describe a high-level vision for how AI can address the specific problem you described in the previous section. It should follow directly from the gaps and opportunities you identified above.

At this stage, do not include any technical details (e.g., models, tools, platforms, architectures, datasets, etc.). We'll spend the next few weeks on the technical system design.

For now, focus on the AI vision. Assuming technical feasibility and successful development, sketch the overall shape of your AI solution with key ideas such as:

- What will it do?
- Who will use it?
- How will it address the problem?
- If successful, how would it change current practice? What other positive impacts would it have?
- If there are existing technical solutions which attempt to address the problem, how will yours differ?

AI Sketch (1-2 paragraphs):

Tracer treats each specimen's journey as one continuous story. Today the laboratory record holds a collection scan, a receipt scan, and a void between them. From the timing patterns the laboratory already produces in ordinary work, Tracer builds a sense of what a normal journey looks like for each kind of specimen, from each collection site, at each hour of the day. It knows that a routine urine leaving a clinic on a weekday afternoon behaves nothing like a spinal fluid drawn in the emergency department at two in the morning. Working from that learned sense of normal, it does three things the laboratory cannot do today. It warns early, raising a flag while a specimen still has time left in its window instead of announcing after the fact that the specimen is late. It makes an informed estimate about the stretch nobody can see, so an alert can point to where the specimen most likely is rather than only reporting that something is wrong. And it weighs each finding by how much harm the failure would cause, so a rare and urgent specimen is escalated differently from a routine one, and every alert reaches the person who can act on it. A specimen that was probably never collected goes to the charge nurse. One stalled in transit

goes to the receiving technologist. One lost on a run between campuses goes to the courier coordinator.

Three groups use it. Specimen processors and technologists work from a live view of everything in flight, ordered by risk. Charge nurses and collection staff receive targeted recovery alerts that name a specific specimen and a specific place to look. Laboratory supervisors and quality staff draw on the accumulating record of what failed and why. If it works, specimen loss becomes something intercepted in progress rather than discovered afterward, and fewer patients face a second draw or a second lumbar puncture for a sample that was already taken once. The laboratory also gains a factual account of its own failure modes in place of anecdotes, which turns process improvement from opinion into evidence. Existing tracking tools answer the question of where a specimen was last seen. Tracer answers where it probably is right now and what is about to go wrong, and it does so without asking the hospital to install any new tracking hardware.